VIRGINIA CECCATELLI

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EDUCATION

McGill University — Montreal, Canada
B.A. Computer Science; Minor in Economics — 2024 – 2026

- Cumulative GPA: **3.76/4.00**, Subject GPA: **3.89/4.00**, Graduated with Distinction (top 25%).
- Graduate-level coursework: Applied Machine Learning, Computer Networks, Compiler Design.
- Girls Who Code (McGill) Vice President; co-developed a Python curriculum; taught weekly advanced-level high school classes.

IE University — Madrid, Spain
B.Sc. International Relations — 2022 – 2024

- GPA: **9.02/10.00**, Dean's List (top 10%); transferred to McGill after completing the second year.
- Honours: Comparative Politics, International Law, Quantitative Methods, International Political Economy, Foreign Aid.
- Research assistant to Karen Nershi: studied statistical patterns in state-backed ransomware activity during electoral cycles.

PUBLICATIONS

Ceccatelli, V., Jeon, Y., & Adelani, D. I. (2026). SpeechJBB: Probing Safety Alignment and Comprehension in Large Audio Language Models under Code-Switched Speech. *Proceedings of **EMNLP** 2026, Main Conference.*

Jeon, Y., Maltais, M., **Ceccatelli, V.**, Ma, M., & Adelani, D. I. (2026). VoxSumm: A Multilingual Corpus of Long-Form Spoken News for Joint Summarization and Translation. *Submitted to **EACL** 2027.*

Ceccatelli, V. & Diop, F. (2026). Securing the Digital Frontier: U.S.–Sub-Saharan Africa Cybersecurity Partnerships in the Shadow of China's Digital Silk Road. *Andalus Committee Policy Paper.*

Research assistance: Nershi, K. (2025). State–cybercriminal alignment in the ransomware ecosystem. *Journal of Cybersecurity.*

Ceccatelli, V. (2024). Sky Diplomacy: The Geopolitical Impact of the Proliferation of Iranian-Russian Military Drone Trade on Global Alliances and Security. *IE International Policy Review.*

RESEARCH EXPERIENCE

UCL S2Lab, Prof. Lorenzo Cavallaro — London, UK (Remote)
AI Security Researcher — June 2026 – Present

- Co-developed and validated a taxonomy of internal LLM uncertainty signals, introducing trajectory-based probes that improved uncertainty estimation across four model families while reducing inference cost relative to sampling-based methods.
- Built a framework to trace binding, data flow, control dependence, and security taint flow in code models, benchmarking ∼2,000 synthetic and real Python programs to estimate latent semantic preservation.
- Verifying causal use of semantic representations via activation patching, and stress-testing robustness under semantics-preserving transformations across 5 obfuscation levels, enabling detection of semantic degradation before it surfaces in outputs.

MILA Quebec AI Institute, Prof. David Adelani — Montreal, Canada
AI Safety Researcher — January 2026 – August 2026

- Led SpeechJBB, the first audio-based code-switching jailbreak dataset for multilingual speech-safety evaluation, exposing weaknesses in SOTA LALM safety alignment; validated findings against multilingual reasoning and ASR capabilities.
- Identified that non-English code-switching increased mean jailbreak success by 28% and reduced refusal by 14.4% relative to monolingual speech; introduced a further pseudo-word obfuscation attack increasing jailbreak success by 34%.
- Created the dataset underpinning Voxsumm, a multilingual corpus for joint summarization and translation of long-form spoken news in low-resource languages, developed in collaboration with Google DeepMind.

McGill Prometheus Lab, Prof. Joseph Vybihal — Montreal, Canada
Computer Vision Researcher — May 2025 – August 2025

- Devised a ground-segmentation and motion-planning system in C++ (U-Net, MobileNetV2), deployed across images, video, and live webcam for autonomous indoor navigation, reaching 0.91 validation IoU on a custom-labelled dataset.
- Selected the deployment checkpoint via safety-driven evaluation, prioritising small-obstacle detection and calibrated confidence on ambiguous surfaces (shadows, reflections) over peak IoU to ensure navigational safety.

PROFESSIONAL EXPERIENCE

WIIT - The Premium Cloud — Düsseldorf, Germany
AI Security Intern — July 2026 – Present

- Collaborating to build an agentic purple-teaming loop that continuously red-teams AI agents and autonomously hardens them.
- Building an automated attacker with defined taxonomy, including prompt injection, tool-parameter manipulation and goal hijaking, as well as safety pillars such as bias and misinformation; producing an evaluation harness to measure success.
- Quantifying detection, bypass and false-positive rates across multiple attack vectors to reduce agent bypass rate.

Wavestone — Munich, Germany
Digital Transformation Intern — June 2024 – July 2024

- Designed an RL system, including environment, state & action spaces, and reward function, to automate data cleaning tasks; reduced records requiring manual review while preserving 96% correction precision against ground truth.
- Improved the overall validity, accuracy and completeness of financial datasets; deployed across German actuarial department.

ADDITIONAL SKILLS

- Technical Skills: Python, C/C++, Java, OCaml, Assembly, Bash, SQL, R, Stata, MATLAB; PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, OpenCV; Git, Docker, Linux; CUDA, SGE/SLURM, LLM APIs; program analysis.
- Languages: Italian (C2). English (C2). German (C2). Spanish (C1). French (C1).